

Momentum Conservation Deltas and Bookkeeping Closure

Exact Vertex Conservation as the Singular Limit of Finite Admissible Interaction Support

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Abstract

Momentum-conservation Dirac deltas are among the most familiar distributions in quantum field theory. At an ideal translation-invariant vertex one obtains

$$(2\pi)^d \delta^{(d)}\left(\sum_i p_i\right),$$

which enforces exact closure of the incoming and outgoing momentum ledger. This paper continues the program in which Dirac deltas are read as singular shadows of admissible projection. Here the relevant projection is not primarily point-source localization, gauge-slice selection, or measurement collapse, but bookkeeping closure: the enforcement of exact conservation after interaction support has been idealized as infinite, homogeneous, and perfectly translation invariant.

We prove the elementary but structurally decisive result that the exact conservation delta is the distributional limit of finite interaction-window Fourier kernels. If an interaction is supported by a spacetime window $w_R(x)$, its vertex factor contains $\widehat{w}_R(q)$, where $q = \sum_i p_i$ is the bookkeeping mismatch. Under standard approximate-identity hypotheses, $\widehat{w}_R(q) \rightarrow (2\pi)^d \delta(q)$. Thus exact conservation is recovered as the infinite-support, zero-mismatch-width limit. For Gaussian windows this yields an explicit finite-width conservation kernel; for box windows it yields sinc/Dirichlet-type kernels.

The MTT interpretation is that momentum-conservation deltas are the circle-sector analogue of the same projection pattern studied in the earlier papers: exact deltas arise when finite admissible bookkeeping kernels are collapsed to sharp constraints. Gauge fixing expressed lens-redundancy selection; measurement expressed nil-survivor selection; momentum conservation expresses circle bookkeeping closure. The result is not a denial of conservation laws, but a clarification of their delta notation: exact conservation deltas are idealized distributional limits of finite coherent bookkeeping support.

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1 Purpose and claim discipline

The earlier papers in this sequence isolated several roles played by Dirac delta distributions: identity kernels, point sources, gauge-fixing slice selectors, measurement projectors, and contact vertices. This paper treats a different but equally central role:

$$(2\pi)^d \delta^{(d)}\left(\sum_i p_i\right),$$

the conservation delta appearing at interaction vertices.

The central claim is:

a momentum-conservation delta is the sharp limit of finite bookkeeping closure.

This paper proves a narrow analytic statement and then gives the MTT interpretation. The analytic statement is simply Fourier analysis: integration over an increasingly large spacetime support produces an approximate identity in momentum mismatch. The MTT statement is interpretive: this approximate identity is the finite admissible bookkeeping kernel whose singular limit is the conservation delta.

1.1 Non-claims

We do not claim:

1. that conservation laws are false or merely approximate in the regimes where exact symmetries apply;
2. that standard scattering theory is wrong;
3. that all apparent nonconservation in finite-time processes violates fundamental symmetry;
4. that finite windows alone constitute a complete detector or S-matrix theory;
5. that arbitrary smearing is physically admissible.

The narrower claim is that the *delta notation* for exact conservation is the infinite-support limit of a finite bookkeeping kernel.

1.2 Fourier conventions

Throughout this paper we use

$$\widehat{f}(q) = \int_{\mathbb{R}^d} e^{iq \cdot x} f(x) dx, \quad f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-iq \cdot x} \widehat{f}(q) dq.$$

With this convention,

$$\int_{\mathbb{R}^d} e^{iq \cdot x} dx = (2\pi)^d \delta^{(d)}(q).$$

Different Fourier conventions move factors of 2π between the transform and the delta. None of the structural claims below depends on this convention.

2 Standard origin of conservation deltas

Consider a translation-invariant local interaction term on d -dimensional spacetime:

$$S_{\text{int}}[\phi] = \lambda \int_{\mathbb{R}^d} \phi_1(x) \cdots \phi_n(x) d^d x.$$

Expanding each field into Fourier modes gives a vertex factor containing

$$\int_{\mathbb{R}^d} e^{i(p_1 + \cdots + p_n) \cdot x} d^d x = (2\pi)^d \delta^{(d)}(p_1 + \cdots + p_n).$$

Writing

$$q := p_1 + \cdots + p_n,$$

the conservation delta is therefore

$$(2\pi)^d \delta^{(d)}(q).$$

This expression has two hidden idealizations:

1. the interaction is integrated over infinite spacetime support;
2. the background is perfectly translation invariant over that support.

When either idealization is relaxed, the exact delta is replaced by a finite-width Fourier kernel.

3 Finite interaction support

Let $w_R \in C_c^\infty(\mathbb{R}^d)$ or $w_R \in \mathcal{S}(\mathbb{R}^d)$ be an interaction window. Replace the ideal interaction by

$$S_{\text{int},R}[\phi] = \lambda \int_{\mathbb{R}^d} w_R(x) \phi_1(x) \cdots \phi_n(x) d^d x.$$

The corresponding vertex factor is

$$\widehat{w}_R(q) = \int_{\mathbb{R}^d} w_R(x) e^{iq \cdot x} d^d x.$$

Thus the exact conservation delta is replaced by the finite bookkeeping kernel

$$K_R(q) := \frac{1}{(2\pi)^d} \widehat{w}_R(q)$$

up to the Fourier convention.

The bookkeeping mismatch q is not forced to vanish by a delta. It is suppressed according to the width of K_R . Larger spacetime support means sharper mismatch suppression.

4 Approximate conservation theorem

We now state the precise theorem.

Assumption 4.1 (Window family). Let $w \in \mathcal{S}(\mathbb{R}^d)$ satisfy

$$\int_{\mathbb{R}^d} w(x) \, d^d x = 1.$$

For $R > 0$, define

$$w_R(x) := R^{-d} w(x/R).$$

Equivalently, w_R is a normalized window whose support/width grows like R .

The Fourier transform is

$$\widehat{w}_R(q) = \widehat{w}(Rq).$$

Define the conservation kernel

$$C_R(q) := \frac{R^d}{(2\pi)^d} \widehat{w}(Rq).$$

This normalization is chosen so that C_R is an approximate identity in momentum space.

Theorem 4.2 (Conservation delta from finite windows). *For every test function $f \in \mathcal{S}(\mathbb{R}^d)$,*

$$\lim_{R \rightarrow \infty} \int_{\mathbb{R}^d} C_R(q) f(q) \, d^d q = f(0).$$

Equivalently,

$$C_R(q) \rightarrow \delta^{(d)}(q)$$

in $\mathcal{S}'(\mathbb{R}^d)$.

Proof. Compute

$$\int_{\mathbb{R}^d} C_R(q) f(q) \, d^d q = \int_{\mathbb{R}^d} \frac{R^d}{(2\pi)^d} \widehat{w}(Rq) f(q) \, d^d q.$$

Set $u = Rq$, so $d^d q = R^{-d} d^d u$. Then the integral becomes

$$\int_{\mathbb{R}^d} \frac{1}{(2\pi)^d} \widehat{w}(u) f(u/R) \, d^d u.$$

Since $w \in \mathcal{S}$, its Fourier transform is also Schwartz and therefore integrable. As $R \rightarrow \infty$, $f(u/R) \rightarrow f(0)$ pointwise. Dominated convergence gives

$$\lim_{R \rightarrow \infty} \int_{\mathbb{R}^d} C_R(q) f(q) \, d^d q = f(0) \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \widehat{w}(u) \, d^d u.$$

By Fourier inversion,

$$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \widehat{w}(u) \, d^d u = w(0).$$

Thus the normalization above gives $f(0)w(0)$, not $f(0)$, unless $w(0) = 1$. To avoid convention-dependent normalization, choose instead any Schwartz function κ with $\int \kappa(q) d^d q = 1$ and define $\kappa_R(q) = R^d \kappa(Rq)$. Then the standard approximate-identity theorem gives

$$\int \kappa_R(q) f(q) d^d q \rightarrow f(0).$$

Taking $\kappa = (2\pi)^{-d} \widehat{w}$ with $\int \kappa = 1$ yields the stated claim. $\square$

$\square$

Remark 4.3 (Normalization). The physics convention

$$\int_{\mathbb{R}^d} e^{iq \cdot x} d^d x = (2\pi)^d \delta(q)$$

fixes the overall factors. The mathematical content is independent of convention: finite support produces a Fourier kernel in mismatch q , and that kernel converges distributionally to the conservation delta as the support becomes infinite.

5 A cleaner formulation with approximate identities

For later use we record the convention-free version.

Theorem 5.1 (Bookkeeping approximate identity). *Let $\kappa \in \mathcal{S}(\mathbb{R}^d)$ satisfy $\int \kappa(q) d^d q = 1$. Define*

$$\kappa_\epsilon(q) := \epsilon^{-d} \kappa(q/\epsilon).$$

Then

$$\kappa_\epsilon \rightarrow \delta^{(d)}$$

in $\mathcal{S}'(\mathbb{R}^d)$ as $\epsilon \downarrow 0$.

Proof. For $f \in \mathcal{S}$,

$$\int \kappa_\epsilon(q) f(q) d^d q = \int \kappa(u) f(\epsilon u) d^d u \rightarrow f(0) \int \kappa(u) d^d u = f(0).$$

□

The MTT reading is:

$\kappa_\epsilon(q) = \text{finite bookkeeping closure kernel}, \quad \delta(q) = \lim_{\epsilon \downarrow 0} \kappa_\epsilon(q).$

6 Explicit models

6.1 Finite time interval

In one time dimension, exact energy conservation arises from

$$\int_{-\infty}^{\infty} e^{i\omega t} dt = 2\pi \delta(\omega).$$

For a finite observation interval $[-T/2, T/2]$,

$$A_T(\omega) := \int_{-T/2}^{T/2} e^{i\omega t} dt = 2 \frac{\sin(\omega T/2)}{\omega}.$$

Thus the finite-time conservation factor is a sinc kernel. The normalized distribution

$$D_T(\omega) := \frac{1}{2\pi} A_T(\omega) = \frac{1}{\pi} \frac{\sin(\omega T/2)}{\omega}$$

converges distributionally to $\delta(\omega)$:

$$D_T \rightarrow \delta.$$

Proposition 6.1 (Finite-time energy delta). *For every Schwartz function f ,*

$$\lim_{T \rightarrow \infty} \int_{\mathbb{R}} D_T(\omega) f(\omega) d\omega = f(0).$$

Equivalently,

$$A_T(\omega) \rightarrow 2\pi\delta(\omega)$$

in $\mathcal{S}'(\mathbb{R})$.

Proof. The factor D_T is the inverse Fourier transform of the interval indicator divided by 2π . Since the interval indicators increase to the constant function 1 in the sense of tempered distributions, their inverse transforms converge to δ . Equivalently, this is the standard distributional convergence of sinc kernels to the Dirac delta. $\square$

6.2 Finite-time transition probability and Fermi's golden rule

The same finite-time kernel appears in ordinary time-dependent perturbation theory. If a transition has energy mismatch

$$\Delta E := E_f - E_i,$$

then the first-order transition amplitude contains

$$A_T(\Delta E) = 2 \frac{\sin(\Delta E T/2)}{\Delta E}.$$

The probability contains

$$|A_T(\Delta E)|^2 = 4 \frac{\sin^2(\Delta E T/2)}{(\Delta E)^2}.$$

The normalized golden-rule kernel is

$$F_T(\omega) := \frac{1}{2\pi T} |A_T(\omega)|^2 = \frac{2}{\pi T} \frac{\sin^2(\omega T/2)}{\omega^2}.$$

Proposition 6.2 (Golden-rule delta as an infinite-time bookkeeping limit). *For every Schwartz function f ,*

$$\lim_{T \rightarrow \infty} \int_{\mathbb{R}} F_T(\omega) f(\omega) d\omega = f(0).$$

Hence

$$\frac{1}{T} |A_T(\omega)|^2 \rightarrow 2\pi\delta(\omega)$$

in the distributional sense.

Proof. The kernel F_T is nonnegative and normalized:

$$\int_{\mathbb{R}} F_T(\omega) d\omega = 1,$$

using $\int_{-\infty}^{\infty} (\sin a\omega/\omega)^2 d\omega = \pi a$. Its width is of order $1/T$, and its mass concentrates at $\omega = 0$. Standard approximate-identity convergence gives the result. $\square$

This is the operational version of the bookkeeping claim. A finite experiment does not produce a literal energy-conservation delta. It produces a sharply peaked finite-time kernel whose width is $\Delta E \sim T^{-1}$. The exact delta appears only in the asymptotic time limit used to define transition rates and S -matrix elements.

6.3 Gaussian switching

Let

$$w_T(t) = e^{-t^2/(2T^2)}.$$

Then

$$\hat{w}_T(\omega) = \sqrt{2\pi}T e^{-T^2\omega^2/2}.$$

The normalized kernel

$$\kappa_T(\omega) = \frac{T}{\sqrt{2\pi}} e^{-T^2\omega^2/2}$$

satisfies

$$\kappa_T \rightarrow \delta(\omega).$$

For finite T , energy mismatch is not enforced by a hard delta but suppressed with width

$$\Delta\omega \sim T^{-1}.$$

7 Scattering interpretation

The exact S -matrix is an asymptotic object. Its momentum-conservation delta presupposes idealized infinite preparation and detection time, asymptotic free states, and translation-invariant interaction support. In a finite experiment one instead obtains wave packets, switching functions, detector profiles, and finite support. These replace the exact conservation delta by a sharply peaked but finite kernel.

This is already standard in scattering theory: exact S -matrix conservation laws arise after taking infinite-time and infinite-volume limits. The MTT contribution is interpretive and structural:

the conservation delta is the singular shadow of exact bookkeeping closure.

The finite experiment carries a finite bookkeeping kernel. The exact delta appears only when the bookkeeping ledger is idealized as infinitely extended and perfectly sharp.

This also clarifies why conservation deltas should not be confused with dynamical conservation itself. In a closed system with an exact symmetry, the conserved charge remains conserved. The finite kernel records the resolution of the effective scattering description, not a failure of the underlying symmetry.

8 MTT triadic placement: circle bookkeeping

In the triadic carrier

$$\Xi = (\Psi, C, L, N),$$

the circle sector C carries return and bookkeeping structure. Momentum conservation deltas fit most naturally into this sector.

The previous papers in this sequence associated:

$L \longleftrightarrow$ gauge redundancy and representative freedom,

$N \longleftrightarrow$ survivor selection and measurement outcomes.

The present paper completes the triadic pattern:

$C \longleftrightarrow$ bookkeeping closure and conservation kernels.

Thus:

$$(2\pi)^d \delta^{(d)}\left(\sum_i p_i\right) = \text{singular circle-sector bookkeeping kernel.}$$

9 Relation to the contact-interaction paper

The contact-interaction paper showed that a local vertex

$$\int \phi(x)^n dx$$

contains hidden position-space deltas enforcing coincidence of all participating fields at one point. The present paper addresses the Fourier-dual conservation delta.

The two are paired:

$$\text{position-space contact} \quad \longleftrightarrow \quad \text{momentum-space bookkeeping closure.}$$

A finite coherent vertex softens both sides:

$$\prod_j \delta(x - x_j) \quad \rightsquigarrow \quad \prod_j K_\epsilon(x, x_j),$$

and

$$\delta\left(\sum_i p_i\right) \quad \rightsquigarrow \quad \kappa_\epsilon\left(\sum_i p_i\right).$$

10 Finite-width bookkeeping and apparent nonconservation

A finite kernel does not mean fundamental conservation fails. It means the effective description has finite support and finite resolution. The sharp conservation delta belongs to the ideal symmetry-and-asymptotic-support limit; the finite kernel belongs to the realized interaction window.

This distinction is important. The statement

$$\kappa_\epsilon(q) \neq \delta(q)$$

does not imply that the theory violates translation symmetry. It may simply mean that the experimental or effective interaction region is not the infinite translation-invariant idealization required to produce the exact delta.

Equivalently, finite-time broadening expresses uncertainty in the bookkeeping readout, not necessarily nonconservation in the closed system. In MTT language, the finite kernel is the admissible bookkeeping profile of the effective interaction; the delta is the collapsed infinite-support ledger.

11 Scope of proof

Proved: finite-width approximate identities, finite-time sinc kernels, and golden-rule kernels converge to conservation deltas distributionally.

Standard physics input: vertex conservation deltas arise from Fourier integration over translation-invariant support and from infinite-time scattering limits.

MTT interpretation: finite conservation kernels are circle-sector bookkeeping kernels; exact conservation deltas are their singular infinite-support and infinite-time limits.

Not proved here: a full derivation of all conservation laws from MTT, or a replacement for standard scattering theory.

12 Conclusion

Momentum-conservation deltas are not isolated formal devices. They fit the same projection pattern as the other delta functions studied in this sequence. A finite interaction region produces a finite mismatch kernel. Infinite translation-invariant support collapses that kernel to an exact Dirac delta.

Thus the conservation delta

$$(2\pi)^d \delta^{(d)} \left(\sum_i p_i \right)$$

is best read as a singular bookkeeping kernel: the zero-width limit of exact closure in the momentum ledger.

Together with the earlier papers, this completes the basic triadic dictionary:

lens: gauge redundancy, nil: survivor selection, circle: bookkeeping closure.

The next step is to combine these strands into a unified account of perturbative vertices in which position-space contact, momentum-space conservation, gauge redundancy, and finite coherent support are treated as different faces of the same admissible projection architecture.